Enrolment Number										

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Winter, 2026

B.Tech 4th Semester Examination

Surveying & Geomatics – Theory

Course Code: BCE23402T

Answer all the questions

SECTION 1

Q1. Choose the correct option: **Marks: 10 x 1 =10**

1. Which instrument is used for measuring horizontal and vertical angles?
   a) Dumpy level b) Prismatic compass c) Theodolite d) Plane table
2. The process of determining elevations of points is known as:
   a) Traversing b) Levelling c) Ranging d) Triangulation
3. Ranging in surveying means:
   a) Plotting maps b) Measuring angles c) Fixing intermediate points on a straight line d) Measuring contours
4. Which curve is introduced to provide gradual change in curvature?
   a) Simple curve b) Reverse curve c) Transition curve d) Vertical curve
5. EDM stands for:
   a) Electronic Data Measurement b) Electromagnetic Distance Mapping c) Electronic Distance Measurement d) Electrical Distance Meter
6. The satellite component of GPS is called:
   a) User segment b) Space segment c) Control segment d) Ground segment
7. The overlap provided between successive aerial photographs along the flight direction is called:
   a) Side overlap b) End lap c) Relief displacement d) Tilt displacement
8. Which platform is most commonly used in remote sensing?
   a) Ship b) Aircraft only c) Satellite d) Compass
9. In plane table surveying, the instrument used for sighting is:
   a) Clinometers b) Alidade c) Sextant d) Pantograph
10. Contour lines connect points of equal:
    a) Distance b) Pressure c) Elevation d) Curvature

SECTION 2

Q2. Answer any ONE

a) Explain the principles of surveying. Name the different types of maps and state their respective uses in chaining. Also mention the different obstacles encountered in chaining. (8)

OR

b) Name the different types of chain available for chain surveying stating their designated lengths? Define the concept of whole circle bearing and quadrantal bearing. Also state the major difference between whole circle bearing and quadrantal bearing. (8)

Q3. Answer any ONE

a) Name the different components of a theodolite and explain the measurement of horizontal angles by method of repetition. (8)

OR

b) Discuss triangulation surveying system and well conditioned triangle. Explain signals and towers used in triangulation survey. (8)

Q4. Answer any ONE

a) Two tangents AB & BC intersect at a point B at a chainage of 150.50m. Calculate all the necessary data for setting out a circular curve of radius 100.00m and angle of intersection 160° by the method of offsets from the long chord. (8)

OR

b) Discuss transition curves and vertical curves used in highway and railway alignment. (8)

Q5. Answer any ONE

a) Explain the principle and working of Electronic Distance Measurement instruments. (8)

OR

b) Describe Total Station surveying procedure, applications, advantages and errors associated with Total Station surveys.

Q6. Answer any ONE

a) Write about the application of Remote Sensing in civil engineering field. Explain the basic principle of ~~civil engineering~~. Remote sensing (8)

OR

b) An aerial survey is to be conducted over an area measuring 12 km × 18 km. The aerial photographs are of size 23 cm × 23 cm with a scale of 1:20,000. Longitudinal overlap is 60% and side overlap is 30%. Determine the approximate number of photographs required. (8)

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